

MALMO, Sweden

WMO: 026350

Lat: 55.5715N

Lon: 13.0708E

Elev: 21

StdP: 101.07

Time Zone: 1.00 (EUC)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.1	-6.7	-12.0	1.3	-7.8	-9.6	1.7	-5.6	11.2	4.8	10.1	4.4	1.6	60	0.551

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.3	26.6	18.8	25.0	18.4	23.3	17.5	20.3	24.4	19.3	23.0	18.4	21.9	3.6	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.7	13.6	22.0	17.8	12.8	21.1	16.9	12.1	20.3	58.4	24.5	54.9	23.2	52.1	21.9	24.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.7	7.7	6.8	-12.9	29.4	3.9	1.6	-15.8	30.5	-18.1	31.5	-20.3	32.4	-23.1	33.5		
			-13.1	22.4	3.8	1.2	-15.9	23.2	-18.1	23.9	-20.3	24.6	-23.1	25.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.0	0.9	1.0	3.1	7.8	12.2	15.5	17.8	17.5	14.2	9.6	5.7	2.6
	DBStd	6.88	3.88	3.56	3.55	3.21	3.06	2.58	2.34	2.30	2.53	3.24	3.27	3.86
	HDD10.0	1251	283	252	215	82	12	1	0	0	2	46	130	228
	HDD18.3	3444	542	486	473	318	192	91	39	43	125	271	378	487
	CDD10.0	898	0	0	0	14	80	166	242	233	128	33	2	0
	CDD18.3	49	0	0	0	0	1	6	22	18	1	0	0	0
	CDH23.3	345	0	0	0	1	12	59	177	93	3	0	0	0
	CDH26.7	41	0	0	0	0	1	7	24	9	0	0	0	0
Wind	WSAvg	3.2	3.7	3.5	3.5	3.4	3.2	3.1	2.9	2.8	3.0	3.0	3.3	3.5
Precipitation	PrecAvg	684	52	39	37	39	48	63	67	79	64	63	69	63
	PrecMax	901	99	89	80	65	114	132	208	221	155	125	156	134
	PrecMin	502	0	8	9	13	14	0	3	12	24	21	15	28
	PrecStd	99	27	16	20	16	20	28	45	55	35	26	30	25
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.1	9.0	13.7	20.7	25.0	27.5	29.0	27.8	23.7	17.8	13.2	10.1
		MCWB	8.0	7.2	9.1	13.8	17.6	18.3	20.2	19.6	17.8	14.5	12.0	9.2
	2%	DB	7.6	7.7	11.1	18.0	22.6	25.2	27.1	25.8	21.6	15.9	11.8	9.2
		MCWB	6.6	6.3	7.5	12.0	15.7	17.9	19.2	18.8	16.7	13.8	10.9	8.3
	5%	DB	6.7	6.7	9.1	15.7	20.6	22.8	25.4	24.0	19.8	14.8	10.7	8.2
		MCWB	5.8	5.6	6.8	10.4	14.4	16.6	18.7	18.1	16.0	13.2	9.7	7.4
	10%	DB	5.8	5.7	7.7	13.7	18.5	20.9	23.6	22.3	18.4	13.8	9.9	7.3
		MCWB	5.0	4.7	6.0	9.4	13.3	15.7	17.9	17.6	15.1	12.3	8.9	6.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.2	7.8	9.7	14.7	18.7	20.6	21.9	21.5	19.1	15.4	12.3	9.7
		MCDB	8.8	8.5	12.1	19.8	23.3	25.8	27.4	25.6	22.2	16.6	12.7	10.0
	2%	WB	6.9	6.6	8.1	12.4	16.7	18.5	20.5	20.1	17.4	14.2	10.7	8.5
		MCDB	7.3	7.4	9.9	16.9	21.3	22.9	24.9	23.6	20.6	15.4	11.3	9.0
	5%	WB	6.0	5.7	7.2	11.0	15.2	17.3	19.6	19.0	16.4	13.4	9.8	7.5
		MCDB	6.6	6.6	8.8	14.7	19.1	21.5	23.6	22.3	19.0	14.6	10.5	8.1
	10%	WB	5.0	4.8	6.2	9.7	13.8	16.3	18.6	18.2	15.6	12.6	8.9	6.6
		MCDB	5.8	5.6	7.7	12.8	17.6	20.1	22.3	21.2	17.8	13.7	9.6	7.3
Mean Daily Temperature Range	5% DB	MDBR	3.8	4.1	6.1	8.5	9.1	8.3	8.3	7.9	7.1	5.6	4.0	3.7
		MCDBR	3.5	4.4	8.1	12.2	12.8	12.1	12.7	10.8	9.1	6.1	3.9	3.6
		MCWBR	3.5	3.5	5.5	7.2	7.1	6.2	6.0	5.2	5.2	4.4	3.5	3.6
	5% WB	MCDBR	3.5	4.2	7.4	10.5	10.8	10.3	10.5	8.9	8.3	5.5	3.7	3.7
		MCWBR	3.6	3.5	5.4	6.6	6.7	6.0	5.8	5.1	5.1	4.3	3.5	3.7
Clear-Sky Solar Irradiance	taub	0.315	0.338	0.360	0.393	0.390	0.384	0.405	0.407	0.375	0.358	0.339	0.306	
	taud	2.285	2.304	2.297	2.271	2.340	2.381	2.341	2.346	2.416	2.395	2.331	2.258	
	Ebn at Noon	594	725	800	825	852	861	832	804	782	695	560	507	
	Edh at Noon	55	78	98	116	115	112	114	107	88	71	53	45	
All-Sky Solar Radiation	RadAvg	0.45	1.06	2.44	4.19	5.39	5.74	5.33	4.29	3.04	1.49	0.57	0.30	
	RadStd	0.09	0.17	0.27	0.51	0.54	0.44	0.65	0.31	0.30	0.22	0.10	0.03	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (60 neighbors)	+0.41	N/A	N/A	N/A	N/A	N/A	N/A	-168	+62	N/A

Nomenclature: See separate page